Chapter 29 — Offline-First Architecture

> **Part IV — Offline-First & Synchronization** · Complexity: **Complex** (≥40 interview questions)
>
> **Prerequisites:** Ch 3 (SSOT / `networkBoundResource` — the read foundation), Ch 12 (caching), Ch 15 (idempotency — safe writes), Ch 6 (UI state), Ch 68 (connectivity, previewed). This chapter is the **umbrella** for Part IV; it introduces the write path (outbox, Ch 31), sync (Ch 30/32), and conflicts (Ch 33), which the next chapters detail.
>
> **Reference surface in StreamCart:** browsing the catalog, editing the cart, and saving wishlist/watch-progress **fully offline**, syncing seamlessly when connectivity returns.

29.1 Problem Definition and Real-World Context

"Offline-first" means the app is **fully usable without a network** and treats the network as an **enhancement**, not a requirement. The local store is authoritative for reads and accepts writes immediately; synchronization happens in the background. This inverts the naive "online-first" model where the UI blocks on the network and is useless on a train, in an elevator, or on flaky café Wi-Fi. Naive apps fail because:

- **The network is the source of truth.** Every screen fetches on demand; no connectivity = spinners and errors; rotation/process death refetch. (Ch 3 established the fix: Room as SSOT.)
- **Writes require connectivity.** "Add to cart" / "like" / "save draft" fail or hang offline; users lose work. Offline-first demands writes **succeed locally immediately** and sync later.
- **No coherent sync model.** Ad-hoc "refresh" buttons, no background sync, no queue for offline writes, no conflict handling → data loss and inconsistency.
- **Poor offline UX.** No indication of what's cached/pending/stale; the user can't tell if their change "took."

The architecture composes patterns from across the book into a coherent system: **SSOT reads** (Ch 3), **optimistic local writes** recorded in an **outbox** (Ch 31), **background sync** via WorkManager (Ch 30) using **idempotent** mutations (Ch 15) and **delta sync** (Ch 32), with **conflict resolution** (Ch 33) and **connectivity awareness** (Ch 68). This chapter defines the end-to-end offline-first design; later Part IV chapters go deep on each piece.

Three practical use cases

1. **E-commerce (StreamCart Shop).** Browse catalog, edit cart, add to wishlist offline; cart/wishlist changes queue and sync on reconnect — no lost actions.
2. **OTT (Watch).** Watch progress, "my list," and downloads work offline; progress syncs across devices when back online.
3. **Productivity/Social (Social).** Compose posts/comments/drafts offline; they appear immediately (optimistic) and publish on reconnect (outbox), with conflict handling if edited elsewhere.

Why this is architectural

Offline-first defines the app's **fundamental data-flow model** (local-authoritative reads + queued writes + background sync), the **consistency guarantees** (eventual consistency, conflict policy), and how **every feature** behaves without a network — the single most impactful architectural decision for mobile UX and reliability.

29.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	Reads serve from the local store (SSOT); work fully offline (Ch 3).
F2	Writes succeed locally immediately (optimistic) and are queued for sync.
F3	Queued writes sync in the background when connectivity returns (Ch 30/31).
F4	Mutations are idempotent so retries/replays don't duplicate (Ch 15).
F5	Remote changes sync down (full or delta) and merge with local (Ch 32).
F6	Conflicts between local and remote are resolved by policy (Ch 33).
F7	The UI communicates sync/pending/stale/offline state.

Non-functional requirements

Attribute	Policy
Availability	App works offline; network is an enhancement.
Consistency	Eventual consistency; defined conflict policy; no silent data loss.
Reliability	No lost writes (durable outbox); survives process death.
UX	Instant local feedback; clear sync status; no blocking spinners for cached data.
Battery/Network	Batched, constrained, delta sync; not constant polling.
Scalability	Works as entities/screens grow; per-entity sync.

Scope

- **In scope:** the offline-first *architecture* — read path (SSOT), write path (optimistic + outbox), sync orchestration, consistency model, UX of sync state, how the pieces compose.
- **Out of scope (deep-dived in Part IV):** WorkManager internals (Ch 30), outbox mechanics (Ch 31), delta sync (Ch 32), conflict resolution (Ch 33), connectivity monitoring (Ch 68).

Assumptions & constraints

- Room (Ch 3), WorkManager (Ch 30), idempotency keys (Ch 15). Backend supports sync (timestamps/versions, delta endpoints). Eventual consistency is acceptable (not a hard real-time/transactional requirement). `minSdk 24`.

29.3 System Design Fundamentals

- **Local store is authoritative for the UI.** The UI reads only from Room (Ch 3); the network updates Room, which re-emits. The user never waits on the network to see data.
- **Optimistic writes.** A write updates the local store **immediately** (the UI reflects it at once) and records a **pending operation** (outbox). The user perceives instant success; the actual server write happens asynchronously.
- **The outbox pattern (Ch 31).** Local writes are durably recorded as pending mutations (with idempotency keys, Ch 15) and replayed to the server by a background worker until acknowledged. This guarantees **no lost writes** across process death / offline periods.
- **Background sync (Ch 30).** A WorkManager job (constrained to connectivity, Ch 68) drains the outbox (push local changes) and pulls remote changes (delta, Ch 32), reconciling into Room.
- **Idempotency (Ch 15).** Because delivery is at-least-once (retries/replays), every mutation carries an idempotency key so the server applies it exactly once.
- **Conflict resolution (Ch 33).** When local and remote diverge (edited on two devices), a policy (last-writer-wins, merge, or user-resolved) decides the outcome, using versions/timestamps.
- **Eventual consistency.** Offline-first trades strong consistency for availability: devices converge over time. The UI must represent **pending/ syncing/synced/conflict** states honestly.
- **Sync status as first-class state.** Each entity/operation has a sync status (synced, pending, syncing, failed/conflict) surfaced to the UI (Ch 6).

Core insight: **make the local store authoritative, accept writes optimistically into a durable outbox, and reconcile with the server via idempotent background sync and a conflict policy** — the network becomes an asynchronous enhancement, and the app is always usable.

29.4 High-Level Architecture


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Components & ownership

- **Room (SSOT)** holds entities **and** their sync status; the UI reads only here.
- **Outbox** durably records pending local writes (Ch 31).
- **SyncWorker** (WorkManager, Ch 30) pushes the outbox (idempotent) and pulls deltas (Ch 32), resolving conflicts (Ch 33) into Room.
- **Connectivity** (Ch 68) triggers/constrains sync.
- **ViewModel** observes Room + sync status; writes optimistically.

Direction: reads/writes go to Room; writes also enqueue to the outbox; the background sync engine reconciles Room ↔ server.

29.5 Detailed Component Design

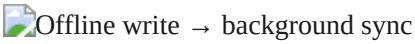


Component	Responsibility	Ch
`Entity` with `syncStatus` / `version` / `updatedAt`	SSOT + sync metadata	3/33
`OfflineRepository`	SSOT reads + optimistic writes + enqueue	3/31
`PendingMutation` (outbox)	durable queued write + idempotency key	31/15
`SyncEngine` / `SyncWorker`	push/pull/reconcile	30/32
conflict resolver	merge policy	33
connectivity	trigger/constraint	68

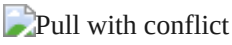
The three sync phases: push (drain outbox → server, mark synced on ack), **pull** (fetch remote changes since last sync → merge), **resolve** (when local pending + remote changed the same entity → conflict policy, Ch 33).

29.6 End-to-End Data Flow

Offline write → background sync



Pull with conflict



29.7 Complete Kotlin Implementation

Package: `com.streamcart.sync` and feature `:data:*`. (Each sub-pattern is detailed in Ch 30–33; here is the composed skeleton.)

29.7.1 Entity with sync metadata

```
enum class SyncStatus { SYNCED, PENDING, SYNCING, FAILED, CONFLICT }

@Entity(tableName = "wishlist")
data class WishlistEntity(
    @PrimaryKey val id: String,              // client-generated UUID for offline creates
    val productId: String,
    val updatedAt: Long,
    val version: Long = 0,                   // for conflict detection (Ch 33)
    val syncStatus: SyncStatus = SyncStatus.PENDING,
    val deleted: Boolean = false,            // soft delete so deletes sync too
)
```

29.7.2 Offline repository: SSOT read + optimistic write + enqueue

```
class WishlistRepositoryImpl @Inject constructor(
    private val dao: WishlistDao,
    private val outbox: OutboxDao,                // Ch 31
    private val syncScheduler: SyncScheduler,     // Ch 30
    private val clock: Clock,
) : WishlistRepository {

    override fun observe(): Flow<List<WishlistItem>> =
        dao.observeActive().map { list -> list.map { it.toDomain() } }   // reads are offline-
first (Ch 3)

    override suspend fun add(productId: String) {
        val entity = WishlistEntity(
            id = UUID.randomUUID().toString(),         // client id -> works offline
            productId = productId, updatedAt = clock.now(), syncStatus = SyncStatus.PENDING,
        )
        dao.upsert(entity)                             // 1) optimistic local write (UI updates
now)
        outbox.enqueue(PendingMutation(               // 2) durable queued write (Ch 31)
            idempotencyKey = UUID.randomUUID().toString(),   // Ch 15
            type = "wishlist.add", entityId = entity.id, payload = json(entity),
        ))
        syncScheduler.requestSync()                    // 3) nudge background sync (Ch 30)
    }

    override suspend fun remove(id: String) {
        dao.markDeleted(id, clock.now())               // soft delete, PENDING
        outbox.enqueue(PendingMutation(UUID.randomUUID().toString(), "wishlist.remove", id, ""))
        syncScheduler.requestSync()
    }
}
```

29.7.3 Sync engine (push → pull → resolve) — orchestration

```
class SyncEngine @Inject constructor(
    private val outbox: OutboxDao,
    private val api: SyncApi,
    private val dao: WishlistDao,
    private val resolver: ConflictResolver,          // Ch 33
    private val syncState: SyncStateStore,           // lastSyncToken (Ch 32)
) {
    suspend fun sync(): Result<Unit> = runCatching {
        push()     // drain outbox (idempotent, Ch 15)
        pull()     // delta (Ch 32) + resolve conflicts (Ch 33)
    }

    private suspend fun push() {
        outbox.pending().forEach { m ->
            dao.setStatus(m.entityId, SyncStatus.SYNCING)
            when (val r = api.apply(m.type, m.idempotencyKey, m.payload)) {   // server dedups
by key
                is Outcome.Success -> { dao.applyServerAck(m.entityId, r.data);
outbox.remove(m.idempotencyKey) }
                is Outcome.Failure -> if (!r.error.isRetryable()) { dao.setStatus(m.entityId,
SyncStatus.FAILED); outbox.remove(m.idempotencyKey) }
                                      // retryable -> leave in outbox; WorkManager retries (Ch
30)
            }
        }
    }

    private suspend fun pull() {
        val since = syncState.lastToken()
        val delta = api.changesSince(since)                    // Ch 32
        delta.items.forEach { remote ->
            val local = dao.findById(remote.id)
            val merged = resolver.resolve(local, remote)       // Ch 33
            dao.upsert(merged)
        }
        syncState.setLastToken(delta.nextToken)
    }
}
```

29.7.4 ViewModel exposes data + sync status (Ch 6)

```
data class WishlistUiState(
    val items: List<WishlistItem>,
    val isOffline: Boolean,
    val pendingCount: Int,          // "2 changes will sync"
    val isSyncing: Boolean,
)
// combine(dao.observeActive(), connectivity.state, outbox.pendingCount(), syncState.isSyncing)
-> UiState
```

29.7.5 Module structure

```
:data:wishlist   WishlistRepositoryImpl (optimistic + enqueue)
:core:database   WishlistEntity (+syncStatus/version), OutboxDao (Ch 31)
:sync            SyncEngine, SyncWorker (Ch 30), SyncScheduler, SyncStateStore (Ch 32)
:core:connectivity  connectivity state (Ch 68)
# ConflictResolver -> Ch 33
```

29.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Local write during an in-flight push	lost/overwritten change	version/`updatedAt` + enqueue new mutation; server dedups (Ch 15)
Push and pull overlap	inconsistent merge	sync in a single transaction/phase order (push then pull); or per-entity locking
Two writes to same entity offline	ordering	outbox is ordered (FIFO per entity); apply in order
Remote + local change same entity	conflict	detect via version; resolve (Ch 33)
Process death mid-sync	partial sync	durable outbox + `lastSyncToken`; resume idempotently (Ch 15/31)
Duplicate sync runs	double apply	idempotent mutations + `enqueueUniqueWork` (Ch 30)
Optimistic UI vs server rejection	stale UI	on non-retryable failure, mark FAILED/rollback; surface to user

- **Idempotency (Ch 15) + a durable ordered outbox (Ch 31) + versioned entities (Ch 33) together make sync safe** under process death, retries, and concurrent edits — the three pillars of correct offline-first.
- **Phase ordering (push before pull)** reduces conflicts: send your pending changes first, then reconcile what came back.

29.9 Failure Scenarios and Recovery

Failure	Detection	Response	UX
Offline	connectivity (Ch 68)	reads from Room; writes queue	fully usable; "will sync"
Reconnect	connectivity	WorkManager drains outbox + pulls	silent sync; status updates
Push rejected (validation)	non-retryable error	mark FAILED; rollback or flag; notify	"couldn't save X"
Conflict (edited elsewhere)	version mismatch	resolve by policy (Ch 33)	merged / user prompted
Process death mid-sync	durable state	resume from outbox + token (idempotent)	seamless
Partial sync	per-item results	retry failed items; keep succeeded	eventual convergence
Duplicate on retry	idempotency key	server dedups (Ch 15)	one effect
Stale local data	pull delta	reconcile; show "updated"	fresh
Outbox grows (long offline)	backlog	bounded; batch sync; compact redundant ops	drains on reconnect

 29.9 Failure Scenarios and Recovery

> **No silent data loss:** a write that can't sync is kept (outbox) and retried; a permanent rejection is surfaced to the user, never dropped. Offline-first's promise is that the user's work is safe.

29.10 Scaling the Implementation

- **Level 1.** Offline reads only: Room as SSOT + `networkBoundResource` (Ch 3).
- **Level 2.** Add optimistic writes + a durable outbox (Ch 31) + WorkManager sync (Ch 30) with idempotency (Ch 15).
- **Level 3 — production.** Delta sync (Ch 32); conflict resolution (Ch 33); connectivity-aware sync (Ch 68); sync-status UX; soft deletes; per-entity sync.
- **Level 4 — large.** Partial/selective sync (only what's needed), paging offline data (Ch 13), background periodic + event-driven sync, backpressure on large outboxes, encrypted offline data (Ch 24).
- **Level 5 — enterprise.** CRDTs or operational transforms for real-time collaborative merge, multi-device convergence guarantees, offline analytics, sync observability/SLAs, conflict audit trails.

Avoid overengineering: not every app needs full offline writes. Many only need offline *reads* (Ch 3). Add the write path (outbox/sync/conflicts) when users must act offline (cart, drafts, field apps) — it's significant complexity.

29.11 Alternative Approaches and Trade-offs

Dimension	Option A	Option B	When
Model	Offline-first (local authoritative)	Online-first (network authoritative)	Offline-first for usability/unreliable networks; online-first for always-connected/strongly-consistent needs
Reads	SSOT (Room)	fetch-on-demand	SSOT for offline/consistency; fetch for purely-remote non-cached
Writes	Optimistic + outbox	pessimistic (block on server)	Optimistic for UX/offline; pessimistic when the user must see server confirmation (payments — but use idempotency + status)
Sync trigger	Event + periodic + on-connect	manual refresh only	Automatic for seamlessness; manual for user-controlled/costly sync
Consistency	Eventual	strong	Eventual for offline-first; strong needs connectivity/transactions
Conflict	LWW / merge / user (Ch 33)	server-wins always	Depends on data; user-resolved for important edits
Sync scope	Delta (Ch 32)	full refresh	Delta at scale; full for small datasets

Offline-first vs online-first: offline-first maximizes availability and UX on mobile but requires the whole sync/outbox/conflict machinery and accepts eventual consistency. Online-first is simpler and strongly consistent but unusable without a network. For consumer mobile, offline-first (at least for reads, often for writes) is usually the right default; strongly-consistent flows (a bank transfer) still use confirmation + idempotency rather than optimistic silence.

Optimistic vs pessimistic writes: optimistic gives instant UX and offline capability but must handle rollback/conflict; pessimistic avoids divergence but blocks on the network (bad offline/UX). Most actions are optimistic; truly critical ones show explicit "processing" with idempotent server confirmation (Ch 15/19).

29.12 Security Considerations

- **Encrypt sensitive offline data** (Ch 24) — offline-first means more data lives on-device longer; a stolen device exposes it unless encrypted (Keystore-wrapped).
- **Clear offline data + outbox on logout** (Ch 22) and scope per account (Ch 42) — pending mutations and cached entities are user data.
- **Idempotency keys** (Ch 15) prevent duplicate side effects from replayed mutations — critical for financial actions synced from the outbox.
- **Server authorizes every synced mutation** — a tampered client could enqueue anything; the backend re-validates authz on each applied operation. Offline-first does not move trust to the client.
- **Conflict resolution must not leak/overwrite** other users' data (Ch 33); server-side checks on merges.
- **Bound offline storage** to avoid filling the device; evict old cached data (Ch 12).
- Client vs backend: the local store is a convenience replica; the backend remains the authoritative, access-controlled source and the arbiter of applied mutations.

29.13 Performance and Resource Optimization

- **Instant reads/writes** (local) = excellent perceived performance; no network on the critical path.
- **Batch + constrain sync** (Ch 30): sync on connectivity/charging, batch outbox pushes, use delta (Ch 32) — not constant polling (battery).
- **Bound the outbox** and **compact** redundant ops (e.g., multiple edits to one field → latest) before pushing.
- **Page offline data** (Ch 13) for large datasets; don't load everything into memory.
- **Avoid sync storms** on reconnect: `enqueueUniqueWork`, backoff (Ch 2/30).
- **Profiling:** sync duration/frequency, outbox size, battery/network from sync, conflict rate, read/write latency.

29.14 Testing Strategy

Test offline behavior, optimistic writes, outbox durability, and sync reconciliation with fakes + in-memory Room.

```
class OfflineWishlistTest {
    @Test fun `add works offline and queues a mutation`() = runTest {
        val repo = repo(online = false)
        repo.add("p1")
        assertEquals(1, dao.observeActive().first().size)       // UI shows it (optimistic)
        assertEquals(1, outbox.pending().size)                  // queued
        assertEquals(SyncStatus.PENDING, dao.findByProduct("p1")!!.syncStatus)
    }

    @Test fun `sync drains outbox and marks synced`() = runTest {
        repo(online = false).add("p1")
        val engine = SyncEngine(outbox, FakeSyncApi(success = true), dao, resolver, syncState)
        engine.sync()
        assertTrue(outbox.pending().isEmpty())
        assertEquals(SyncStatus.SYNCED, dao.findByProduct("p1")!!.syncStatus)
    }

    @Test fun `retryable failure keeps mutation for later`() = runTest {
        repo(online = false).add("p1")
        SyncEngine(outbox, FakeSyncApi(ioError = true), dao, resolver, syncState).sync()
        assertEquals(1, outbox.pending().size)                  // retried next run
    }

    @Test fun `duplicate sync does not double-apply (idempotent)`() = runTest {
        repo(online = false).add("p1")
        val api = CountingSyncApi()
        val engine = SyncEngine(outbox, api, dao, resolver, syncState)
        engine.sync(); engine.sync()
        assertEquals(1, api.appliedKeys.size)                   // server dedups by key (Ch 15)
    }

    @Test fun `pull reconciles remote changes`() = runTest {
        val engine = SyncEngine(outbox, FakeSyncApi(delta = listOf(remote("p2"))), dao,
resolver, syncState)
        engine.sync()
        assertNotNull(dao.findByProduct("p2"))
    }
}
```

Also: conflict scenario (Ch 33), process-death resume (outbox persists), soft-delete sync, connectivity-triggered sync (Ch 30/68), and a full offline→online integration test.

29.15 Production Observability

- **Metrics:** sync success/failure rate, sync latency/frequency, outbox size & drain time, pending-mutation age (how long writes stay unsynced), conflict rate & resolution outcomes, offline-usage rate, write-loss incidents (should be 0).
- **Logs:** `sync{phase, pushed, pulled, conflicts, duration, result}`, `mutation{type, status}` (ids/types, no PII, Ch 28), correlation IDs (Ch 1/84).
- **Debugging:** writes "disappear" → outbox not durable / not retried; duplicates → missing idempotency (Ch 15); data not updating → pull/delta broken (Ch 32); battery drain → over-frequent/unconstrained sync (Ch 30); stuck pending → non-retryable error not surfaced.
- **Alerting:** rising pending-mutation age or sync-failure rate → backend/sync incident.
- **Privacy:** offline data + outbox are user data; redact logs (Ch 28), honor deletion.

29.16 System Design Interview Preparation

> 40+ questions.

Fundamentals

1. *What is offline-first?* The app is fully usable offline; local store is authoritative; network is an enhancement; sync happens in the background.
2. *vs online-first?* Online-first blocks on the network (useless offline); offline-first serves local and syncs later.
3. *Read path?* SSOT — UI reads Room; network updates Room (Ch 3).
4. *Write path?* Optimistic local write + durable outbox (Ch 31) + background sync (Ch 30).
5. *Consistency model?* Eventual consistency with a conflict policy (Ch 33).

Write path / outbox

6. *What's the outbox?* A durable queue of pending local mutations replayed to the server until acked (Ch 31).
7. *Why durable?* Survive process death / long offline; no lost writes.
8. *Why idempotency keys?* At-least-once delivery (retries/replays) → exactly-once effect (Ch 15).
9. *Client-generated ids?* So creates work offline before the server assigns an id.
10. *Soft deletes?* So deletes sync (a hard-deleted row can't be pushed).

Sync engine

11. *Sync phases?* Push (drain outbox), pull (delta), resolve (conflicts).
12. *Why push before pull?* Send local changes first, then reconcile — fewer conflicts.
13. *What triggers sync?* Connectivity return (Ch 68), periodic, and on write (Ch 30).
14. *Delta vs full sync?* Delta (changes since a token) scales; full for small data (Ch 32).

Conflicts

15. *When do conflicts happen?* Local pending + remote changed the same entity.
16. *How detect?* Version/`updatedAt` mismatch (Ch 33).

17. *Resolution strategies?* LWW, field merge, user-resolved (Ch 33).

Concurrency/reliability

18. *Process death mid-sync?* Durable outbox + last-sync token; resume idempotently.
19. *Duplicate sync runs?* Idempotent mutations + `enqueueUniqueWork` (Ch 30).
20. *Two offline edits same entity?* Ordered outbox; apply in order.
21. *No lost writes guarantee?* Durable outbox + retry + surface permanent failures.

UX

22. *Sync status?* Per-entity status (synced/pending/syncing/failed/conflict) surfaced to UI (Ch 6).
23. *Optimistic rollback?* On non-retryable rejection, revert + notify.
24. *Offline indication?* Connectivity state + pending count (Ch 68).

Security

25. *Sensitive offline data?* Encrypt (Ch 24); clear on logout (Ch 22); per-account (Ch 42).
26. *Does offline-first trust the client?* No — server authorizes every applied mutation.
27. *Financial actions offline?* Idempotent + explicit status; not silent optimistic.

Performance/scale

28. *Battery?* Constrained/batched/delta sync (Ch 30/32), not polling.
29. *Large offline datasets?* Paging (Ch 13), selective sync, eviction (Ch 12).
30. *Outbox backlog after long offline?* Bound + compact redundant ops.

Testing

31. *How test offline writes?* Fake offline; assert optimistic + queued.
32. *How test sync?* Fake API; assert drain + status; idempotent double-sync.
33. *How test process-death resume?* Persist outbox; re-run; no dup.

Production debugging

34. *Writes disappear?* Non-durable/non-retried outbox.
35. *Duplicates?* Missing idempotency.
36. *Data not updating?* Broken pull/delta.
37. *Battery drain?* Over-frequent/unconstrained sync.
38. *Stuck pending?* Non-retryable error not surfaced.

Design

39. *When NOT offline-first (writes)?* Always-connected/strong-consistency flows; then offline *reads* only.
40. *CRDTs?* For real-time collaborative merge (advanced, Ch 33).

Design-change challenges

1. **Make the cart editable offline.** Optimistic writes + outbox + idempotent sync.
2. **Support two-device editing.** Versioning + conflict resolution (Ch 33).
3. **Handle week-long offline.** Bounded/compacted outbox; delta sync on reconnect.

4. **Payments offline.** Idempotency + explicit status, not silent optimism (Ch 15).
5. **Real-time collaboration.** CRDT/OT merge; push via WebSocket (Ch 63).

Mock walkthrough (abridged)

Clarify (offline reads only or writes too? consistency? conflicts?) → SSOT reads (Ch 3) → optimistic writes + durable outbox (Ch 31) + idempotency (Ch 15) → WorkManager sync push/pull (Ch 30) + delta (Ch 32) → conflict policy (Ch 33) → sync-status UX → security (encrypt/clear) → eventual consistency trade-offs → testing.

29.17 Practical Exercises

1. **Beginner.** Make a list read offline-first (Room SSOT + `networkBoundResource`, Ch 3). *Eval:* works with no network; refreshes online.
2. **Intermediate.** Add optimistic create/delete with a durable outbox + idempotency key; drain it on reconnect. *Eval:* instant UI; queued; synced; no dup on double-sync.
3. **Advanced.** Add delta pull (Ch 32) + LWW conflict resolution (Ch 33) + sync- status UX. *Eval:* remote changes merge; conflicts resolved; status shown.
4. **Production debugging.** Fix an app that loses offline writes (non-durable outbox) and duplicates on retry (no idempotency). *Eval:* durable + idempotent.
5. **System design challenge.** Design offline-first for a collaborative doc with two-device edits (versioning + conflict policy; consider CRDT). *Eval:* converges; no lost edits.

(Solutions withheld unless requested.)

29.18 Chapter Summary & Reference Sheet

Key decisions: make the **local store authoritative** (SSOT reads, Ch 3); accept **optimistic writes** into a **durable outbox** (Ch 31) with **idempotency keys** (Ch 15); reconcile via **background sync** (WorkManager, Ch 30) that **pushes then pulls** (delta, Ch 32) and **resolves conflicts** (Ch 33); model **eventual consistency** and surface **per-entity sync status** (Ch 6); use **client-generated ids** + **soft deletes** so creates/deletes work offline; **encrypt + clear** offline data (Ch 24/22); the **backend still authorizes** every applied mutation. **No silent data loss** — the user's work is always safe.

Concepts: offline-first vs online-first, SSOT reads, optimistic writes, outbox, idempotency, background sync, delta sync, conflict resolution, eventual consistency, sync status.

Common mistakes: network as source of truth; blocking writes on the network; non-durable outbox (lost writes); no idempotency (duplicates); no conflict policy (lost edits); hard deletes (can't sync); unconstrained sync (battery); no sync-status UX; unencrypted offline data; trusting the client for authz.

Checklist:

- [ ] SSOT reads (Ch 3)
- [ ] optimistic writes
- [ ] durable outbox (Ch 31)

- [ ] idempotency keys (Ch 15)
- [ ] WorkManager sync push→pull (Ch 30)
- [ ] delta sync (Ch 32)
- [ ] conflict policy (Ch 33)
- [ ] client ids + soft deletes
- [ ] sync-status UX
- [ ] encrypt + clear offline data
- [ ] server authorizes mutations.

Key APIs/patterns: Room (SSOT, Ch 3), `networkBoundResource`, `Outbox`/`PendingMutation` (Ch 31), WorkManager (Ch 30), idempotency keys (Ch 15), delta tokens (Ch 32), `ConflictResolver` (Ch 33), `SyncStatus`, connectivity (Ch 68).

References:

- Offline-first (data layer): https://developer.android.com/topic/architecture/data-layer/offline-first
- Guide to app architecture: https://developer.android.com/topic/architecture
- WorkManager (Ch 30): https://developer.android.com/topic/libraries/architecture/workmanager
- (composes) Ch 3 SSOT, Ch 15 Idempotency, Ch 30 WorkManager, Ch 31 Outbox, Ch 32 Delta Sync, Ch 33 Conflicts

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/failure diagrams; composed offline-first skeleton (entity + optimistic repo + sync engine + UI status); concurrency (3 pillars: idempotency/outbox/versioning); failure (no silent loss); 5 levels; offline-vs-online and optimistic-vs-pessimistic trade-offs; security (encrypt/clear/authz); perf (batch/delta/bound); offline + sync tests; observability (pending age); 40 Q&A + 5 challenges. ✅
- **Expand later:** this is the umbrella — WorkManager (Ch 30), outbox (Ch 31), delta sync (Ch 32), conflict resolution (Ch 33) go deep next; CRDTs, real-time (Ch 63).
- **Interview coverage:** exceeds ≥40 bar. **Opens Part IV.**